

Strengthening FAA Academy Training Outcomes Through System Alignment

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Source Referenced: CAMI paper, "Improving FAA Academy Training Outcomes for ATC," by Chanda Sanders, Ph.D., and Braden Tanner, Ph.D., Civil Aerospace Medical Institute (AAM-520).

Source and Author Note

This white paper was prepared by Chris Wilbanks, Vice President, Mission Support. It references the CAMI paper, "Improving FAA Academy Training Outcomes for ATC," as the research basis for the issues discussed. CAMI's paper identifies several challenges in aligning training systems. The connection to APEX represents the author's assessment of how the APEX concept aligns with and may help address those challenges.

Executive Summary

A recent CAMI paper, Improving FAA Academy Training Outcomes for ATC, provides an important and timely framework for how the FAA should think about improving controller training outcomes. The central message is that Academy performance should not be measured only by how quickly students move through training. Speed matters, but speed alone is not equivalent to system efficiency.

True efficiency occurs when the Academy produces graduates who are prepared in ways that align with the expectations, conditions, and training demands they will face upon arrival at field facilities. The Academy is not intended to produce fully certified professional controllers. That responsibility remains with facility training. However, the Academy must prepare students, assess their ability to continue toward certification, and provide the field with a clearer understanding of each trainee's demonstrated readiness.

The CAMI paper highlights a key source of friction in the current training system: the field and the Academy do not always share the same understanding of the Academy's role. In the field, we routinely hear frustration that Academy graduates arrive underprepared. In many cases, that frustration reflects a deeper system issue: a lack of shared expectations about what the Academy is designed to produce, what readiness should mean, and what information should transfer with the trainee.

The paper argues that improving Academy outcomes requires alignment across three areas: performance measurement and expectations, readiness signaling at transition points, and the use of downstream field performance to improve upstream Academy training. Those three areas tie directly to the purpose and value of APEX: creating an enterprise training performance intelligence capability that connects Academy performance, transition readiness, and field outcomes.

The Core Issue: Role Clarity Between the Academy and the Field

The Academy's purpose is not to produce a controller who is ready to perform as a CPC upon arrival at a facility. The Academy's purpose is to provide initial qualification training, assess whether the student has demonstrated the core capability to continue toward certification, and prepare the trainee for successful entry into facility training.

If the field expects Academy graduates to arrive at or near facility-certification readiness, frustration is inevitable. Conversely, if the Academy only communicates a pass/fail outcome without a more meaningful readiness profile, the field is left to rediscover strengths, weaknesses, and support needs after the trainee arrives.

This is the gap the CAMI paper helps define. It is not enough to say a student passed the Academy. The system needs to define what passing represents, what readiness means, what was actually demonstrated, and what the receiving facility should know before training begins.

Three High-Leverage Areas Identified in the CAMI Paper

1. Strengthen Performance Measurement and Expectations

The paper argues that performance measurement must become more structured, consistent, and reliable. Knowledge-based testing is easier to standardize. The harder challenge is performance-based evaluation, where instructors and evaluators must assess judgment, timing, prioritization, workload management, error recovery, and decision-making under pressure.

The paper highlights that these types of evaluations can allow subjectivity when human judgment is not supported by clear, observable performance indicators and calibrated standards. If two instructors evaluate the same performance differently, the system loses confidence in progression decisions and makes it harder to identify why students succeed or struggle.

The paper also points to the need to align lessons, simulations, and evaluations with clear learning objectives. This is especially important because we continue to hear concerns that we may be evaluating skills or performance differently than we instruct them. If that is occurring, it is a major training-design issue. Students should be instructed, practiced, and evaluated against aligned expectations. A disconnect between instruction and evaluation creates confusion, undermines fairness, and weakens the diagnostic value of assessment data.

2. Clarify and Standardize Readiness

The paper argues that the Academy needs to move beyond a simple pass/fail signal. Passing the Academy should not be the only information available to the receiving facility. The field needs a clearer, structured picture of what the trainee demonstrated and what type of support may be needed next.

A stronger readiness signal could include performance in high-workload scenarios, consistency across simulations, ability to prioritize, ability to recover from mistakes, decision-making under pressure, and known areas for continued development. This type of readiness profile would help facilities understand the trainee earlier and reduce the need to diagnose everything from the beginning after arrival.

This is where the field's frustration must be translated into a system improvement opportunity. The answer is not simply to blame the Academy or the field. The answer is to define readiness more clearly, measure it consistently, and communicate it in a way that helps the next stage of training.

3. Use Downstream Field Performance to Improve Upstream Academy Training

The paper emphasizes that the strongest test of Academy preparation occurs after students arrive in the field. If trainees consistently struggle with similar issues once they enter facility training, those patterns should be captured and used to improve Academy training design, simulation scenarios, instructional emphasis, and evaluation methods.

For example, if multiple trainees struggle with workload management, prioritization, timing, or decision-making as traffic complexity increases, that should no longer be anecdotal feedback. It should be structured data that allows the Academy and the agency to understand where the training system is producing gaps and where changes are required.

Without this downstream-to-upstream feedback loop, the FAA risks addressing problems too late in the pipeline. With the feedback loop, recurring field-training challenges can inform earlier interventions and better Academy preparation.

Connection to APEX

The CAMI paper ties directly to the need for APEX. The paper calls for integrated trainee performance data, objective performance tracking, early indicators of difficulty, structured readiness profiles, error trend analysis, and stronger feedback mechanisms between the Academy and field facilities. These are the capabilities APEX is intended to support.

APEX should not be viewed narrowly as a technology project. It is a training-system alignment effort. The purpose is to provide the FAA with better visibility into how students are performing, where they are struggling, what patterns are emerging, and how those patterns connect to field outcomes.

APEX enables the agency to move from anecdotal observations to data-informed decisions. Today, too much of the conversation around Academy performance is shaped by generalized field feedback, individual instructor judgment, or after-the-fact observations. APEX can help create a more objective, enterprise-level view of trainee performance across the full training lifecycle.

Why This Matters

The FAA’s controller workforce challenge cannot be solved by throughput alone. Moving more students through the Academy is necessary, but it will not be sufficient if the system does not clearly understand readiness, consistently measure performance, and use field outcomes to improve earlier training stages.

The better question is not simply, “How many students can we move through the Academy?” The better question is, “Are we producing the right level of readiness, measuring it consistently, and using what happens in the field to continuously improve the Academy?”

The Academy does not need to replicate facility certification. That is not its role. But it does need to produce a clearer, more measurable, and more operationally aligned level of readiness. To do that, the FAA needs a connected training data system that links Academy performance, transition readiness, and downstream field outcomes.

Summary of the Alignment Opportunity

CAMI Paper Identifies	Current System Challenge	APEX Opportunity
Performance measurement must be more consistent.	Instructor interpretation and subjective evaluation can create variability.	Standardized performance data, evaluator calibration support, and trend analysis.
Readiness must be clearly defined and communicated.	Pass/fail outcomes do not provide the field with sufficient information.	Structured readiness profiles that transfer with the trainee.
Training, simulation, and evaluation must align with objectives.	There is concern that instruction and evaluation may not always measure the same thing.	Objective-level mapping between curriculum, simulation events, and evaluation outcomes.
Field performance should improve Academy training.	Downstream struggles are not consistently captured or used to refine training.	Feedback loop connecting field outcomes back to Academy design and intervention.
System visibility is needed across the training lifecycle.	Academy and field training data are fragmented.	Enterprise performance intelligence from Academy entry through field progression.

Conclusion

The CAMI paper provides a strong framework for rethinking Academy modernization. It reinforces that improving outcomes is not only about speed, class size, facilities, or simulation capacity. Those matter, but they must be connected to a broader system that defines readiness, measures performance consistently, and learns from downstream results.

This is the value of APEX. If implemented correctly, APEX can help the FAA improve Academy efficiency, increase consistency, better support trainees and instructors, reduce preventable training delays, strengthen field training, and improve the controller workforce pipeline in ways that support both safety and operational readiness.